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DEPARTMENT OF INFORMATION AND COMMUNICATION TECHNOLOGY
Faculty of Science & Technology
Bangladesh University of Professionals (BUP)
Course Title: **Digital Communication (ICE 3105)** (3rd Year, 1st Semester, January-June, 2026)
Class Test-01

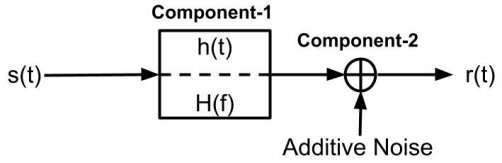
Time: **25 minutes** Full Mark: **20**

Instructions:

1. Answer all the questions.
2. For Questions **1** to **10** (Each contributing 1 mark), put a tick (✓) on only one option.
3. Complete your answers for the questions **11** and **12** (Each contributing 5 mark) within the prescribed space. No Extra page will be provided.

1. Acoustic channels are the most suitable for-
   (a) Space Communication (b) Underwater Communication
   (c) Wired Communication (d) Air Communication
2. Positive value of channel gain in an Additive Noise Channel-
   (a) Attenuates the signal (b) Can do both (a) and (b)
   (c) Amplifies the signal (d) Can do either (a) or (b)
3. In a White Gaussian Noise, the component signals are of
   (a) Equal frequency (b) Equal amplitude
   (c) Equal power (d) Equal noise
4. What is an impulse?
   (a) A signal with unit area (b) A signal having infinity amplitude at $t = 0$
   (c) A Signal having 0 amplitude at $t \neq 0$ (c) All of (a), (b), and (c)
5. Distortion(s) for an Additive Noise Channel-
   (a) Attenuation (b) Modification of Magnitude
   (c) Modification of phase (d) All of (a), (b), and (c)
6. The LTV filter-
   (a) Must be linear (b) Time Variant
   (c) Characterized by impulse and frequency response (d) All of (a), (b), and (c)
7. Distortions for an LTI-filter channel-
   (a) Noise (b) Attenuation
   (c) Modification of signal spectrum (d) All of (a), (b), and (c)
8. The operation for an output from an LTI on Input Signal and Impulse Response is-
   (a) Simple Multiplication (b) Convolution
   (c) Fourier Series (d) Addition
9. What is the correct property of Cumulative Distributive Function (CDF)-
   (a) $cdf(-\infty) = 1$ (b) $cdf(+\infty) = 0$
   (c) $0 \leq cdf(x) \leq 1$ (d) All of (a), (b), and (c)
10. Let, $cdf(4) = 1.2$, $cdf(8) = 4.7$, and $cdf(13) = 9.7$, determine $Pr\{8 \leq x \leq 13\}$
    (a) 5.0 (b) 3.5
    (c) 8.5 (d) 4.0

11. Explain the two components in the following channel. 5



12. Given that, a sample space contains $cdf(x) = \begin{cases} 1 - e^{-x^2}, x \geq 0 \\ 0 \quad\quad\quad\quad , x < 0 \end{cases}$ 5

Determine-

(a) $Pr\{x \leq 5\}$

(b) $\Pr\{x \in [10, 13]\}$

And explain the physical meaning of the obtained values for (a) and (b).